

Lesson 13 — Forever & Stop

Matches LEGO Teacher Portal lesson: "Winning Waterslide".

Objective (I can...): I can use a forever loop and know how to stop it. **CS concept:** Infinite loops and controlled stopping · **Builds on:** Lesson 12 (loop bodies).

AI Connection (2 min): An always-on assistant runs a forever loop, listening until you speak. *Impacts prompt:* "Who does an always-listening speaker help? Could it bother anyone in the house?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Loops model from Lesson 11 — code only.

60-minute flow - Warm-up (8): "Never stop" game — start a chant with no end; ask how to stop it. Introduce a loop that runs until told otherwise. - **Concept + demo (10):** Demo a forever loop (blinking light); stop it with the Stop button, then better, with an event. - **Build/Code (30):** Pairs program a forever loop (constant blink or spin) on the built model, then add a way to stop it (Stop button or event). *Swap at 15 min.* - **Challenge (7):** Use one event to start the forever loop and another to stop it. - **Share + pack-up (5):** Pairs demo starting and stopping. Model stays built.

Pair roles: Driver codes the loop, Navigator owns the stop plan. *Swap at 15 min. (Trio rotates start/loop/stop.)*

Differentiation: *Support* — use the Stop button only. *Extension / fast-finisher* — start and stop entirely with events. *Reteach* — run the forever loop, feel how it never ends, then add the stop together.

Exit check: Pair runs a forever loop and cleanly stops it as planned. **At-home (optional):** Name a machine that "runs forever until turned off"; how would you stop it? ~10 min.
